

MTL106: Probability and Stochastic Processes

Practice Sheet 1 – Solutions

1. Sample Spaces

- (a) Three advertisements (A, B, C) on two platforms (Facebook, Instagram):

$$\Omega = \{(A, F), (A, I), (B, F), (B, I), (C, F), (C, I)\}.$$

- (b) Outcomes of next three tennis matches (W = win, L = lose):

$$\Omega = \{WWW, WWL, WLW, LWW, WLL, LWL, LLW, LLL\}.$$

- (c) Weather prediction for next two days (S, R, C):

$$\Omega = \{S, R, C\} \times \{S, R, C\}.$$

2. σ -fields on $\Omega = \{1, 2, 3, 4\}$

Three σ -fields satisfying $\mathcal{F}_1 \subset \mathcal{F}_2 \subset \mathcal{F}_3$ are:

$$\mathcal{F}_1 = \{\emptyset, \Omega\},$$

$$\mathcal{F}_2 = \{\emptyset, \Omega, \{1, 2\}, \{3, 4\}\},$$

$$\mathcal{F}_3 = \mathcal{P}(\Omega) \text{ (power set).}$$

3. Checking probability function on $\Omega = \{1, 2, 3\}$

Given

$$P(1) = 0.2, \quad P(2) = 0.5, \quad P(3) = 0.3.$$

Clearly,

$$P(i) \geq 0 \quad \text{for all } i,$$

and

$$P(1) + P(2) + P(3) = 0.2 + 0.5 + 0.3 = 1.$$

Hence, P is a valid probability function.

4. Checking probability function on $\Omega = [0, 1]$

Given

$$P(x) = \begin{cases} 2x, & x \in [0, 1/2], \\ 2(1-x), & x \in (1/2, 1]. \end{cases}$$

Since $P(x) \geq 0$ for all $x \in [0, 1]$ and

$$\int_0^{1/2} 2x \, dx + \int_{1/2}^1 2(1-x) \, dx = \frac{1}{4} + \frac{1}{4} = \frac{1}{2},$$

the total probability is not equal to 1.

Hence, P is **not** a probability function.

5. Website advertisement clicks

Out of 2500 visitors:

$$P(1 \text{ click}) = \frac{30}{2500}, \quad P(2 \text{ clicks}) = \frac{15}{2500}, \quad P(3 \text{ clicks}) = \frac{5}{2500}.$$

(a) Probability of clicking on an advertisement:

$$P(\text{click}) = \frac{30 + 15 + 5}{2500} = \frac{50}{2500} = 0.02.$$

(b) Probability of clicking on at least two advertisements:

$$P(\geq 2) = \frac{15 + 5}{2500} = \frac{20}{2500} = 0.008.$$

(c) Probability of not clicking on any advertisement:

$$P(0) = 1 - \frac{50}{2500} = 0.98.$$

6. Leap year with 53 Sundays

A leap year has 366 days = 52 weeks + 2 days. For 53 Sundays, one of the two extra days must be Sunday.

$$P = \frac{2}{7}.$$

7. Proof of identity

$$(A \cap B^c) \cup (A^c \cap B) = A \triangle B.$$

Thus,

$$P(A \triangle B) = P(A) + P(B) - 2P(A \cap B).$$

Hence proved.

8. All 60 students have different birthdays

$$P = \frac{365}{365} \cdot \frac{364}{365} \cdots \frac{306}{365}.$$

9. At least two students share a birthday

$$P(\text{at least one match}) = 1 - P(\text{all different}).$$

10. Electric bulbs problem

There are 4 defective bulbs.

$$P(\text{room is lighted}) = 1 - P(\text{all defective}) = 1 - \frac{\binom{4}{3}}{\binom{10}{3}}.$$